Recitation 11

Colab & Video Summarization

Agenda

1. Google Colab
2. Motion Segmentation
3. Video Summarization

סקר הוראה

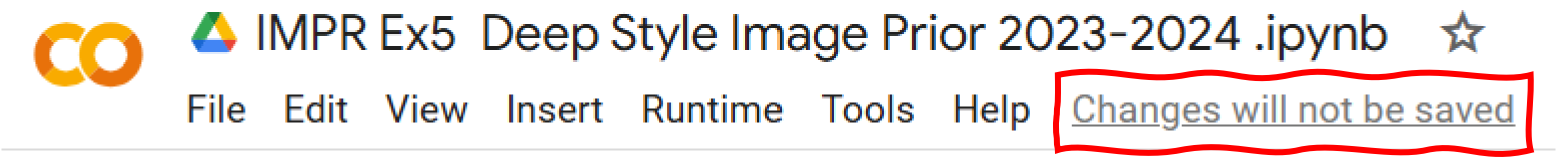
Colab

Motion Segmentation
Video Summarization

Using Colab

The basics:

- Running code in blocks – remember to re-run blocks after change.
- Auto-save – IMPORTANT: if doesn't auto-save check your internet connection.
- Easy to run on GPU.



Using Colab - GPU

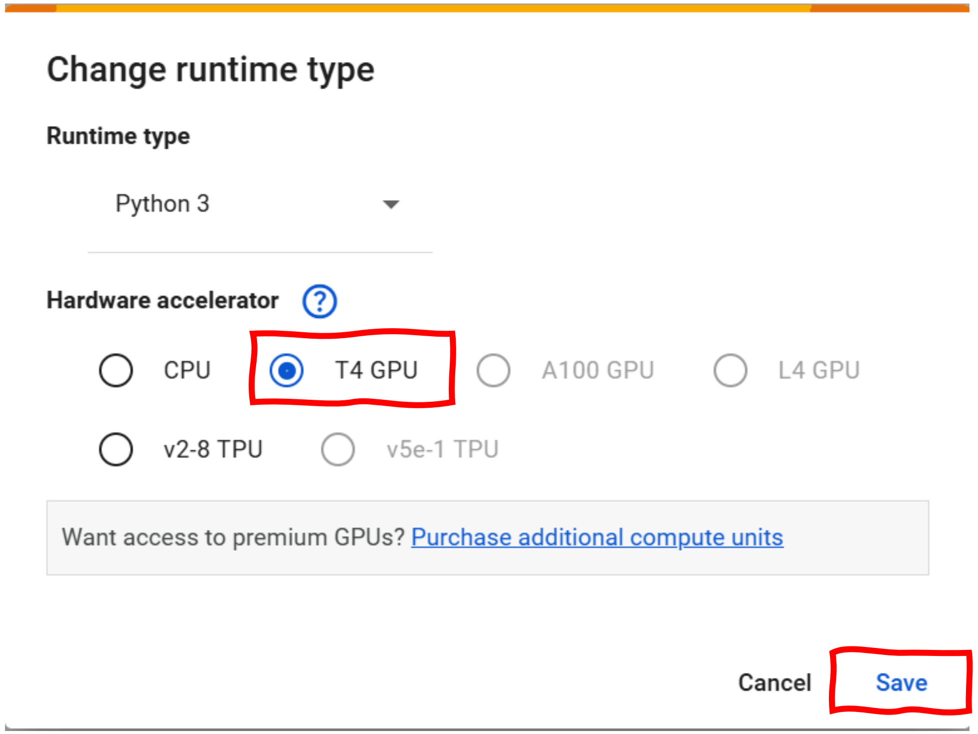
Why GPU? Significantly faster model training and execution.

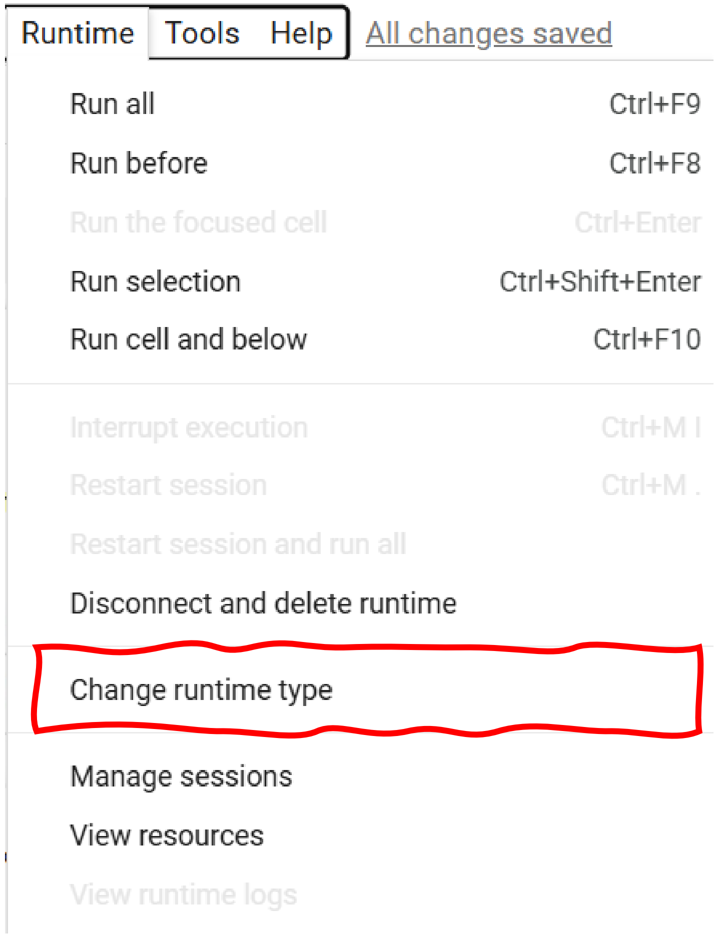
- Google gives each student one hour a day. What can we do? Pay!
  - Begin the exercise early.
  - Use friend’s account.
  - Purchase additional GPU resources.

Using Colab - GPU

How to run on GPU:

"Runtime" → "Change runtime type" → "GPU" → "SAVE"

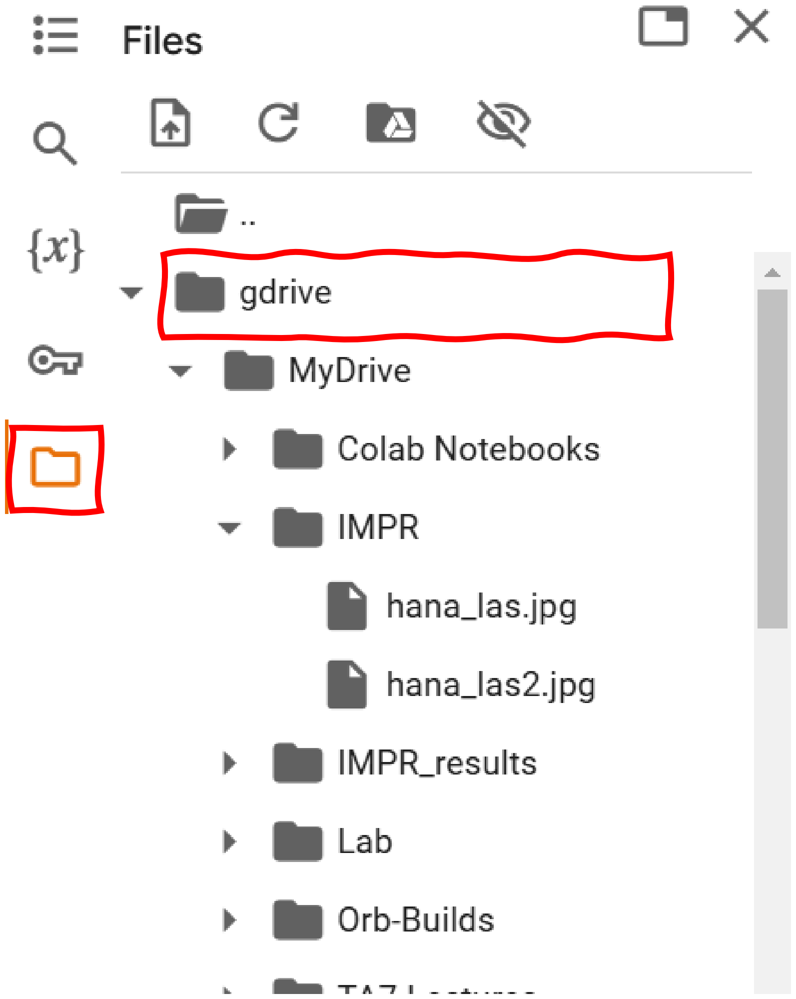




Using Colab

How to mount your Drive and excess files:

```python
from google.colab import drive
drive.mount('/content/gdrive/')
```



Colab

Motion Segmentation

Video Summarization

Motivation

Goals of motion detection:

- Identify moving objects
- Detection of unusual activity patterns
- Computing trajectories of moving objects

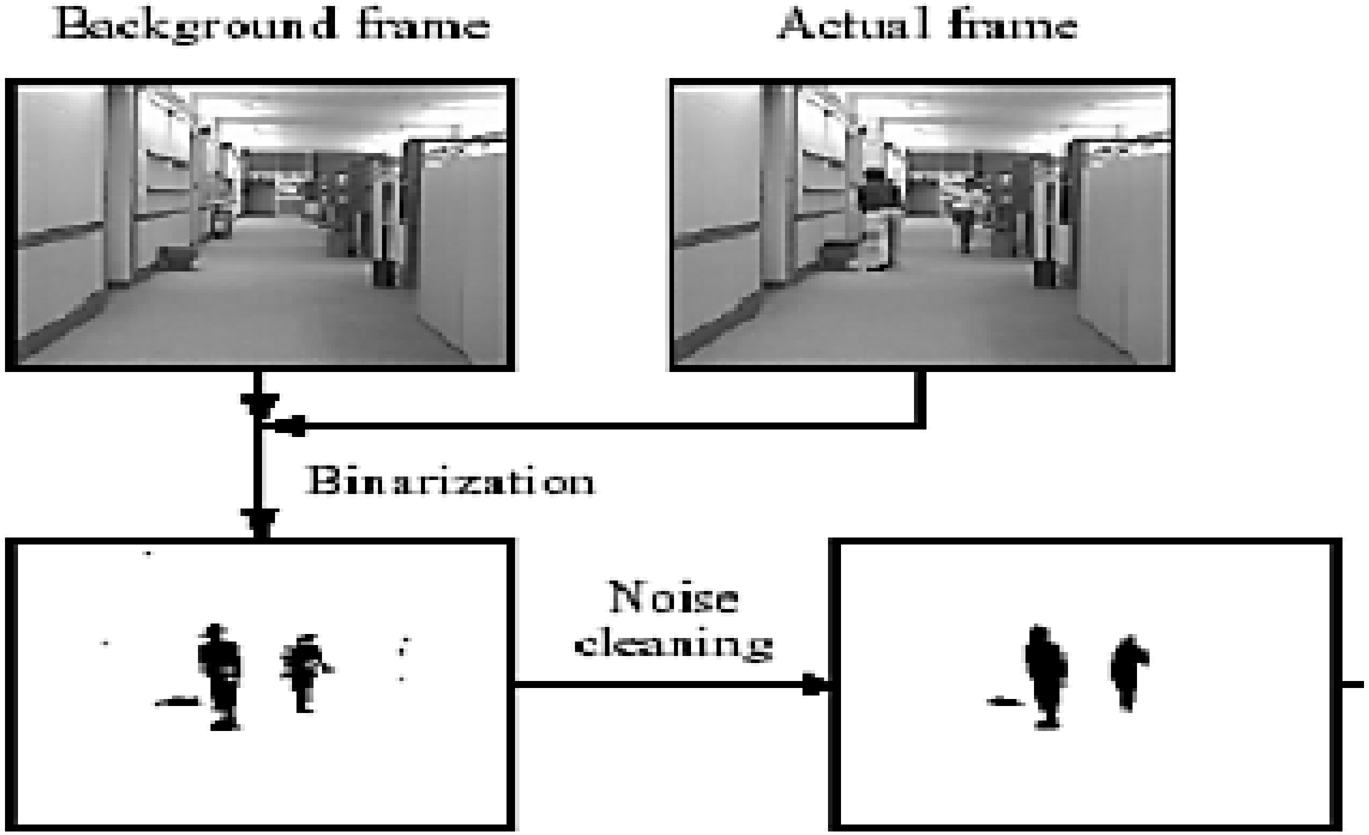
- Applications of motion detection:
- Indoor/outdoor security
- Real time crime detection
- Traffic monitoring
- Many intelligent video analysis systems are based on motion detection.

Background Subtraction

- Uses a reference background image for comparison purposes.
- Current image (containing target object) is compared to reference image pixel by pixel.
- Places where there are differences are detected and classified as moving objects.

Background Subtraction

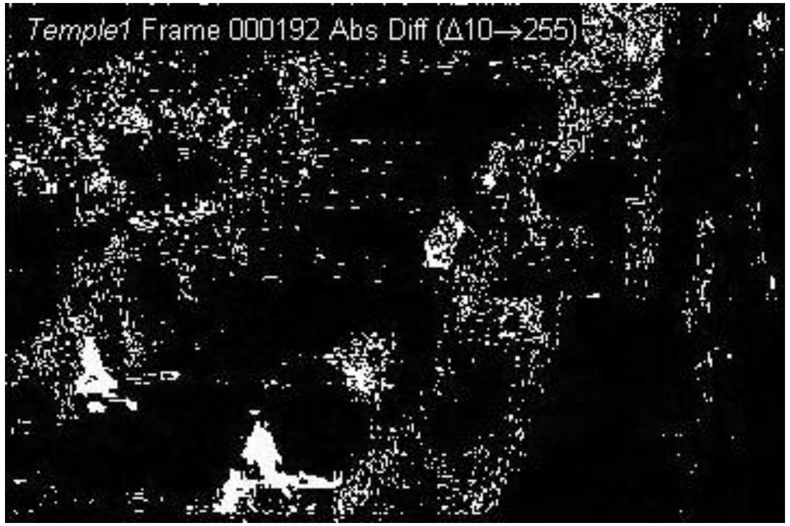
1. Construct a background image B as a median of few images
2. For each actual frame I, classify individual pixels as foreground if $|B - I| > T$ (threshold)
3. Clean noisy pixels



Background Subtraction

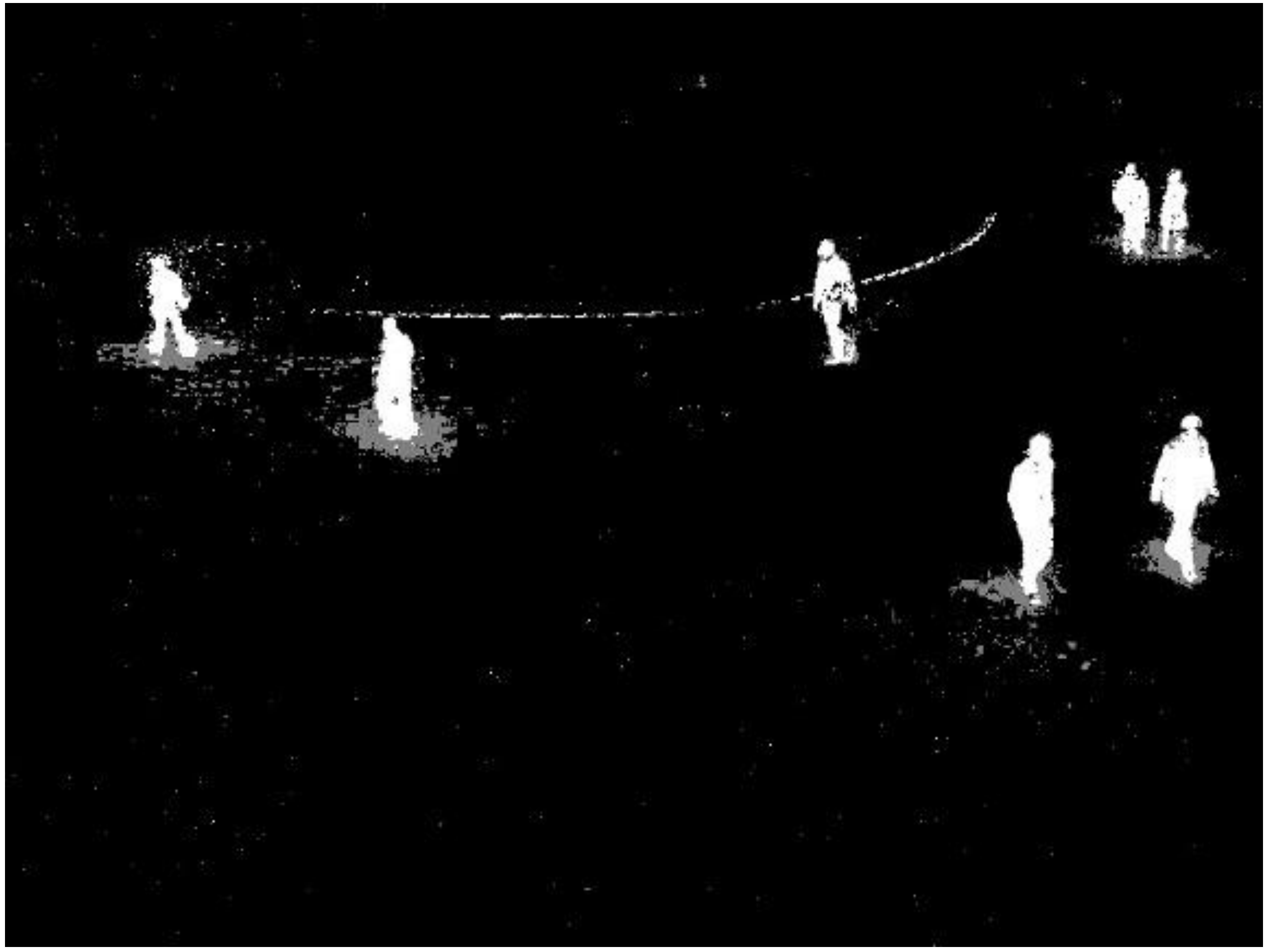
Background Image

Current Image

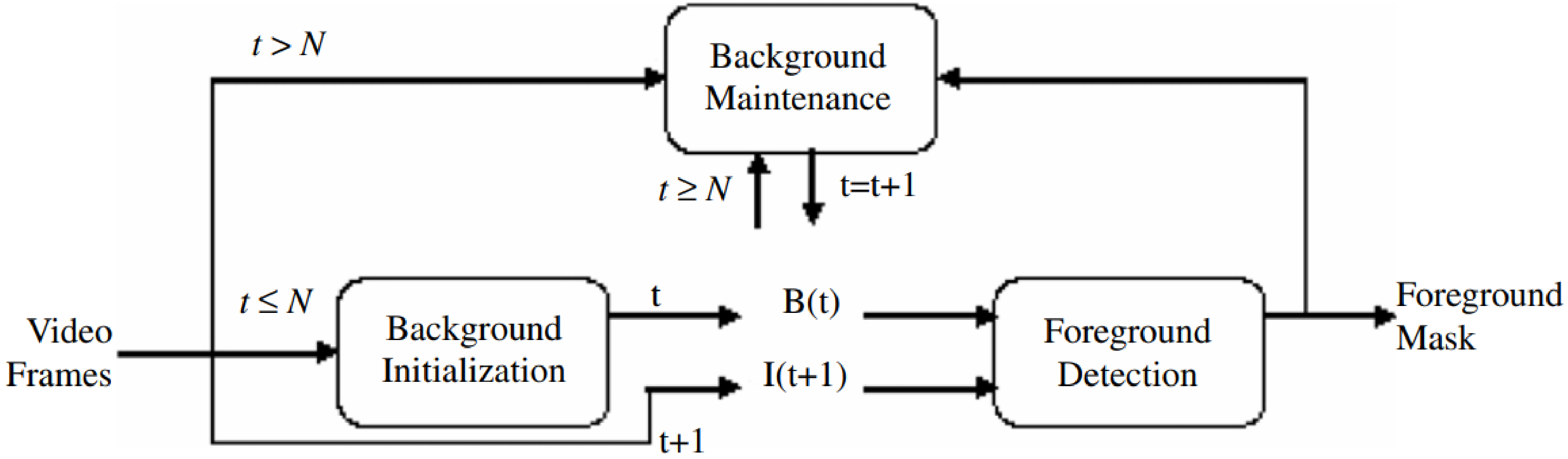


Subtraction + Threshhold 100

Background Subtraction



Background Maintenance



$$B_{t+1}(x,y) = (1-\alpha)\,B_t(x,y) + \alpha I_t(x,y) \quad \textit{if } (x,y) \textit{ is background}$$

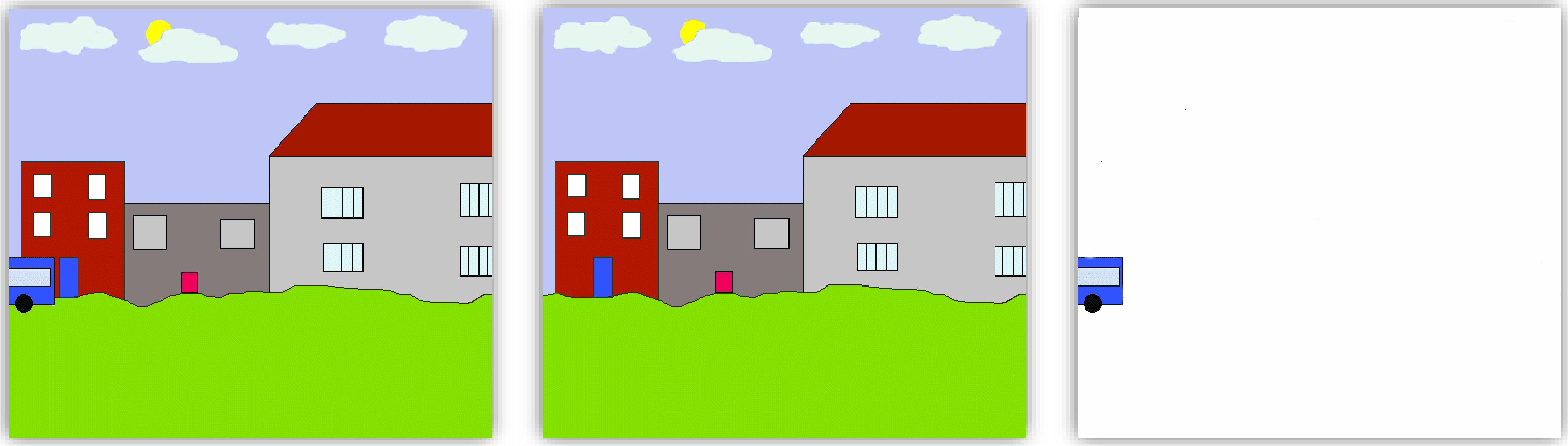
$$B_{t+1}(x,y) = (1-\beta)\,B_t(x,y) + \beta I_t(x,y) \quad \textit{if } (x,y) \textit{ is foreground}$$

$$\beta \ll \alpha$$

Optical Flow can help detect moving objects

Static Scene Object Detection and Tracking

- Model the background and subtract to obtain object mask
- Filter to remove noise
- Group adjacent pixels to obtain objects
- Track objects between frames to develop trajectories



Colab

Motion Segmentation

Video Summarization

Video Summarization

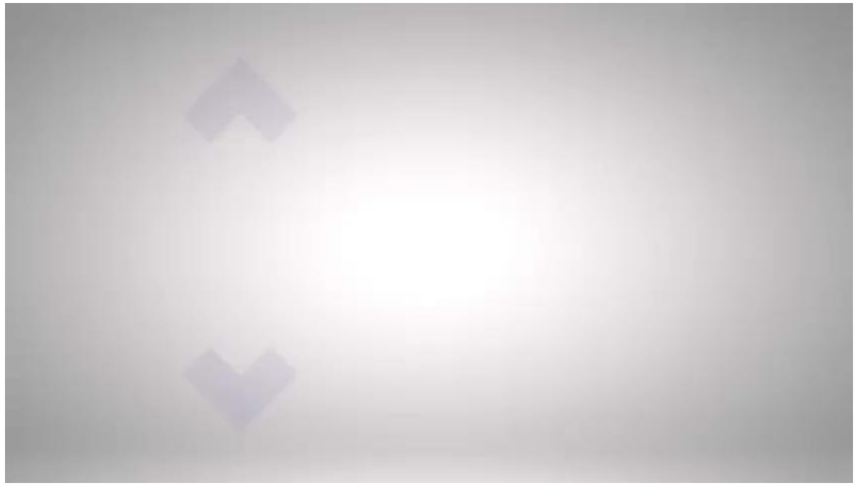
- The digital world contains huge amounts of recorded video (~85%, with ~70% captured by surveillance cameras)
- Video browsing is time-consuming, resulting with most captured video never being watched or examined
- Video summarization helps summarize the content of a long video into a compact representation
    - Enables to quickly review a long video

Video Summarization

We will present two methods

Summarizing a video into a static mosaic image

Summarizing a video into a video synopsis



Using Video Synopsis

Boston bombing (2013): FBI released photos of suspects after 2 days!

Finding Sinwar (2024)

Without Video Synopsis

London Tube Terrorists
July 2005

Dubai Assassination
January 2010

It took <u>weeks</u> to find these events in video archives

Video Indexing Based on Mosaic Representations

Michal Irani and P. Anandan (IEEE 1998)

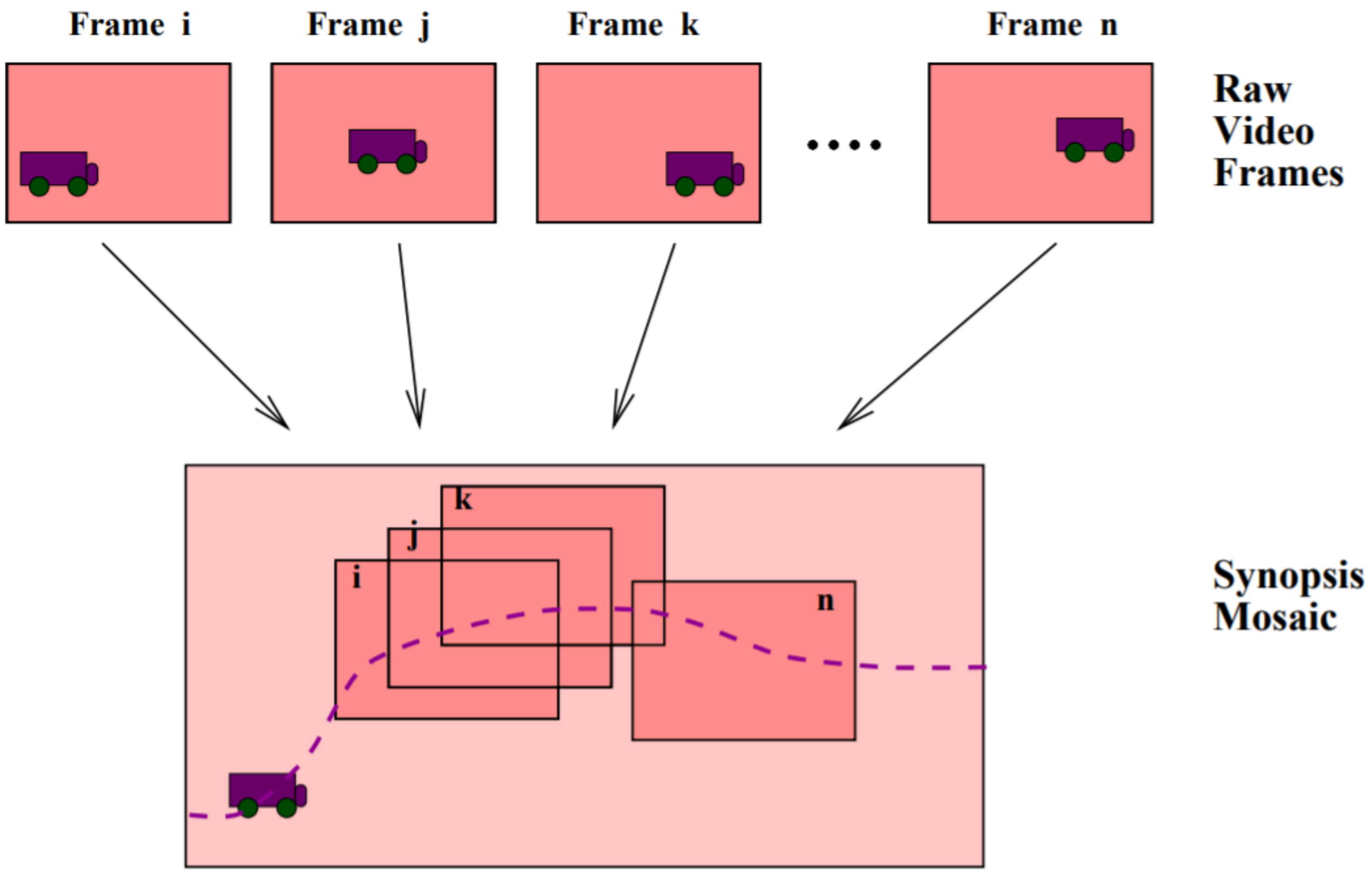
Scene-Based Representation

- Videos contain many frames of the same scene over time
  - High temporal redundancy
- The idea: transform a video from a sequential **frame**-based representation to a **scene**-based representation
- The scene-based representation is based on the three fundamental information components of video:
  - Extended spatial information – appearance of the scene
  - Extended temporal information: motion of moving objects in the scene
  - Geometric information: 3D scene structure, geometric transformation caused by camera motion

Scene-Based Representation

- Extended spatial information
  - Due to camera movement, the static appearance of the scene can be distributed among many frames – no way to see entire scene at once
  - Represented as a panoramic mosaic image, capturing an extended spatial view of the entire scene
- Extended temporal information
  - The evolution of events over time is distributed over frames
  - Represented using the time trajectory of each moving object's center of mass
- Geometric information
  - Relating the different frames to the mosaic coordinate system

From Frames to Scene



Static Background Mosaic

- Align all frames in the scene to a single coordinate system
- Integrate the frames to form a single static mosaic image
    - Remove the moving objects (e.g. temporal average or median)
- Save the geometric transformation relating the different video frames to mosaic coordinates

Static Background Mosaic

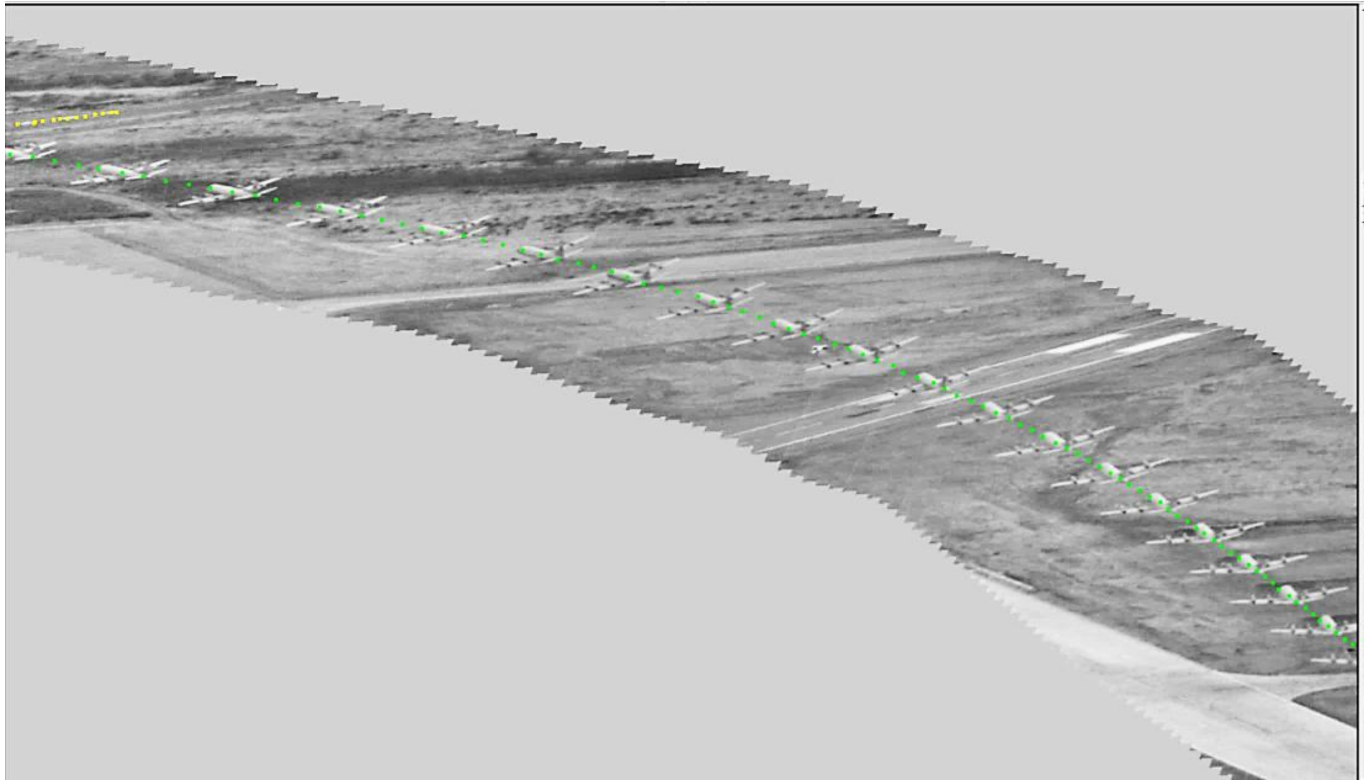
Static Background Mosaic

Synopsis Mosaic

- Moving objects are not represented in the static mosaic
- To provide a summary of the events, a trajectory of each moving object is added on top of the static mosaic
- Creates a visual summary of the entire dynamic foreground event that occurred in the video

Synopsis Mosaic

Synopsis Mosaic



Synopsis Mosaic

Synopsis Mosaic

Nonchronological Video Synopsis and Indexing

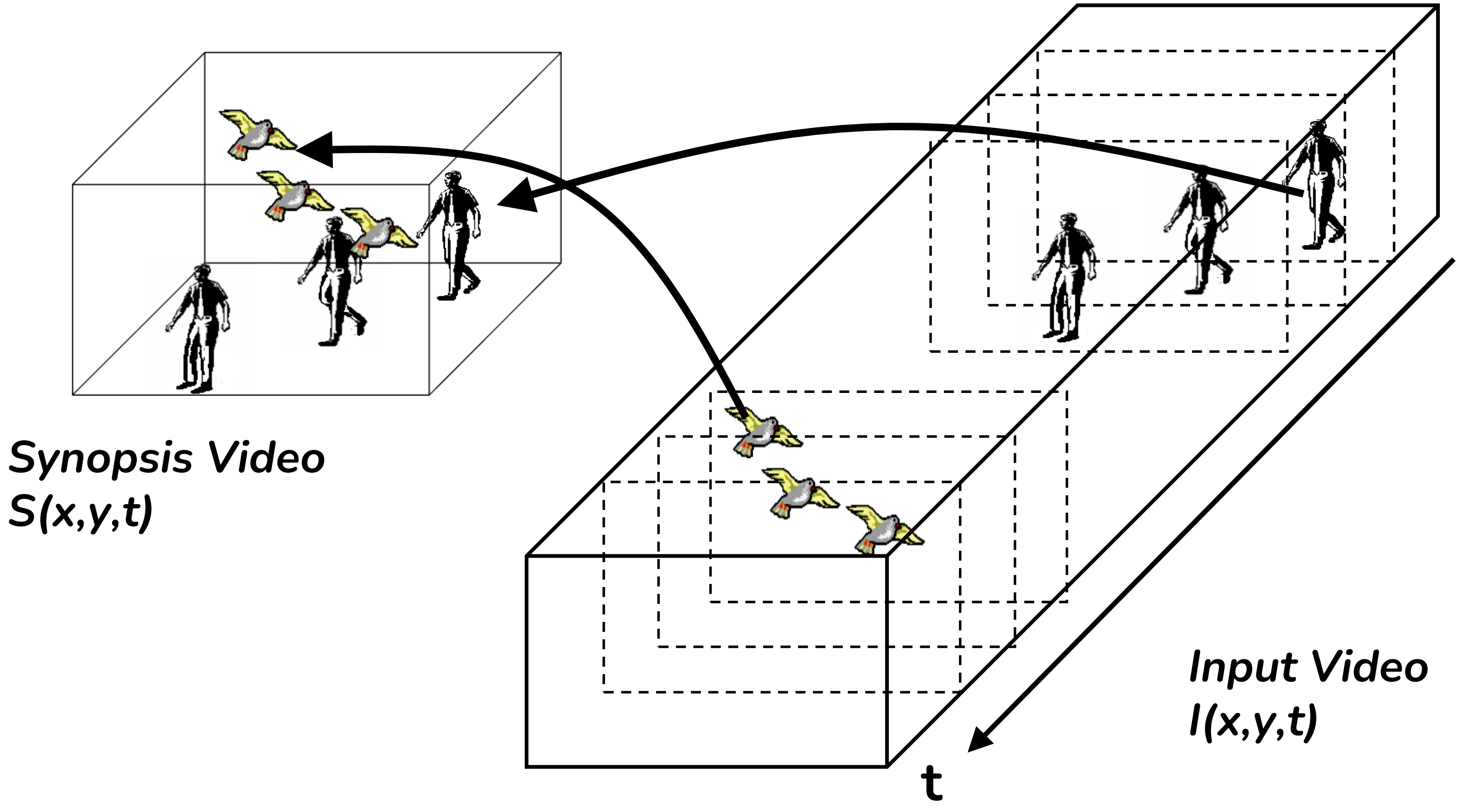
Yael Pritch, Alex Rav-Acha and Shmuel Peleg (PAMI 2008)

Object-Based Summary

- Most previous works suggested **frame**-based summary
  - Detect key frames
  - Show short video segments
  - Adaptive fast forward
- Video synopsis proposed an **object**-based summary
  - Simultaneously show multiple activities that may have originally occurred at different times
  - No fast forward – preserves dynamics

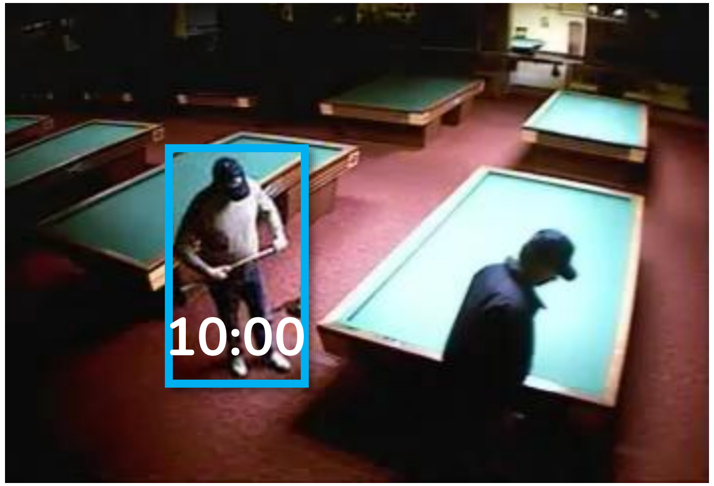
Object-Based Summary

- Main idea: shift objects in time

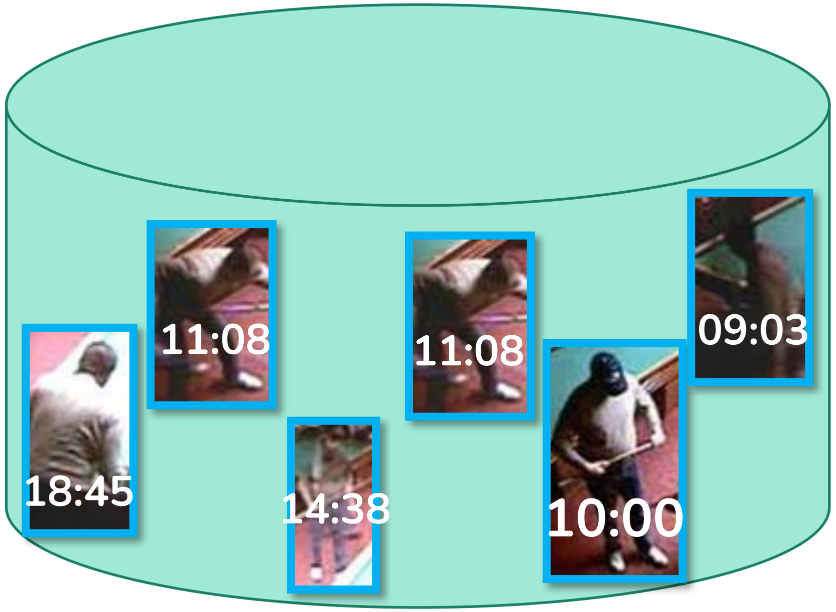


Video Synopsis - Steps

- Two phase process:
  - Online phase: detect and track objects (tubes) and store in database
  - Query phase: according to a given user query ("30 seconds synopsis of the last day") select the relevant objects and stitch them into the synopsis



Original: 9 hours



Database

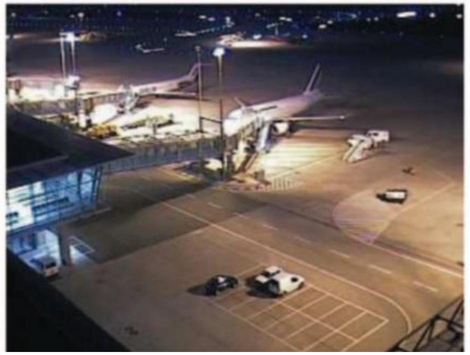
Synopsis: 30 seconds

Video Synopsis - Properties

- Let N frames of an input video be represented in a 3D space-time volume $I(x, y, t)$
  - (x, y) are the spatial coordinates of the pixel and $1 \leq t \leq N$ is the frame number
- The synopsis video $s(x, y, t)$ should have the following properties:
  - S should be substantially shorter than the original video I
  - Maximum "activity" from the original video should appear in the synopsis
  - Dynamics should be preserved (no fast-forward)
  - Visible seams and fragmented objects should be avoided

Object Detection and Segmentation

- We need to identify interesting objects and activates
  - Interesting ≈ moving objects
    - Some motions are not interesting, like clouds and leaves
    - Some interesting objects, like people, can be static
    - Object recognition techniques can be incorporated to support these issues
  - Describe the objects as **tubes** in the 3D space-time volume
- Start with background construction: $B(x, y, t)$
  - Appearance of background can change over time (day-night)
  - Use temporal median over a few minutes before and after frame t
  - In practice: temporal histogram per-pixel

Activity Tube Construction

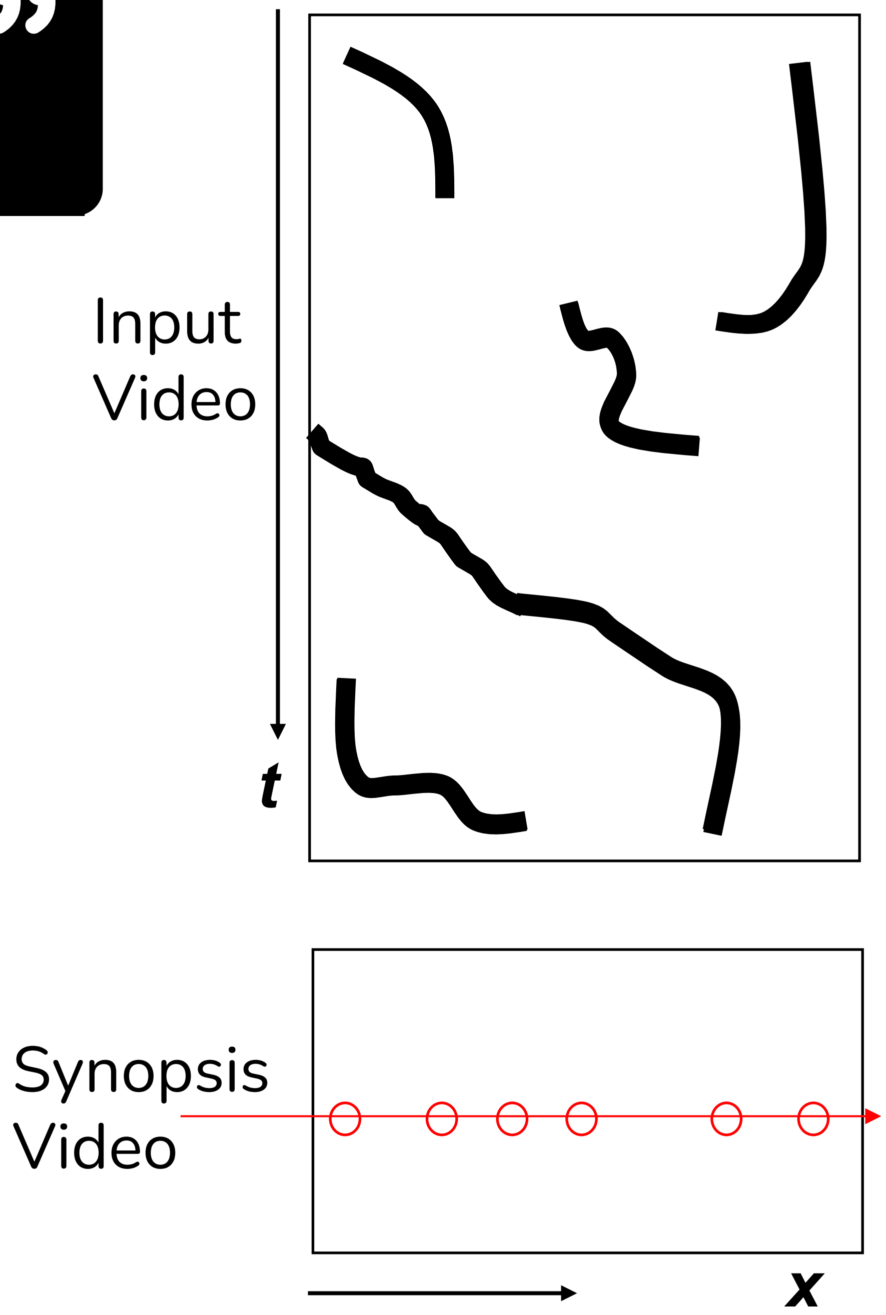
- First step: for each frame I_t, create a labeling function f_t such that $\forall r \in I_t$:
- $$f_t(r) = \begin{cases} 1, & foreground \\ 0, & background \end{cases}$$

Activity Tube Construction

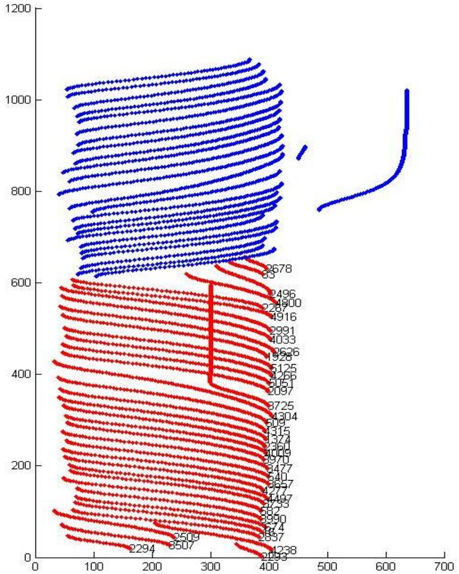
- Labeling function f_t creates blobs
- Overlapping between consecutive blobs used to create 3D $(X \times Y \times T)$ connected components: **activity tubes**
- Each tube b is defined over a finite time segment in the original video: $t_b = [t_b^s, t_b^e]$

Tube “Packing”

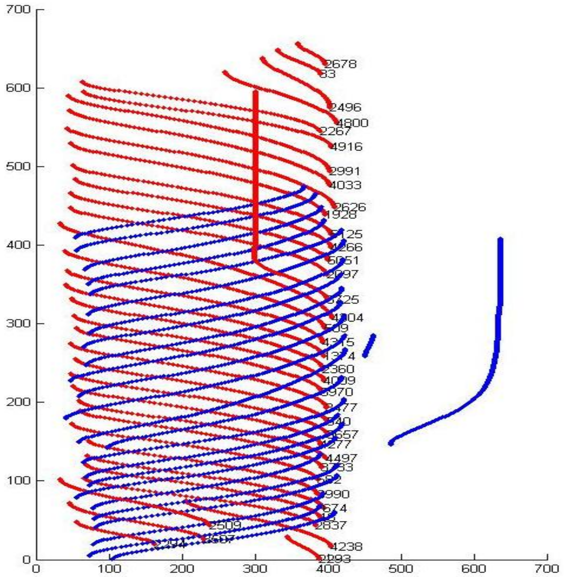
- Find M that enables us to create the optimal time shift for each tube
  - Shortest video
  - Minimum collision



High Collision Cost



Low Collision Cost



Stroboscopic Video

- The length of the synopsis video is lower bounded by the length of the longest tube
  - Problem when very long tubes exists
- Solution: cut the long tube into shorter sub-tubes and display them simultaneously
- Can result with a stroboscopic effect

Stitching the Synopsis

- Some objects can have different lighting then the background
  - Night object at day background
- A seamless stitching method must be added
  - Poisson blending. Pyramid blending works as well.

Stitching the Synopsis

Next week:
Restoration